

Troubleshooting**DANGER****TURN ELECTRICAL AND AIR OFF BEFORE
CLEANING OR REPAIRING MACHINE****DANGER****IMPORTANT**

Troubleshooting should be done by qualified personnel only.

Problem	Cause	Solution
The machine is plugged in but nothing is operating.	There is no power at the plug connection.	Test for voltage at the main disconnect and the circuit breaker.
	There is no secondary voltage from the main transformer.	Test for voltage into the transformer. Test the primary and the secondary fuses.
The machine has power but will not turn on.	The primary fuse is blown.	Replace the fuse.
	The E-Stop switch is not working.	Test the circuit, check the contact block. Replace the switch if it is bad.
	The motor contact relay has failed.	Replace the relay.
The Count Photoeye is not counting	The lens is obstructed or dirty.	Clear the obstruction away or wipe the lens using a soft cloth.
	There is no power to the photoeye.	Check all the electrical connections and the power.
	The photoeye is not set to its maximum setting.	Adjust the photoeye to its maximum setting at the adjustment pot. Full adjustment has been reached when the pot screw clicks while being turned.
The Stack Counts are incorrect.	The product pieces are overlapped coming from the ambient cooler.	Separate the product before it reaches the Counter Stacker.
	The Count Photoeye is dirty.	Clean the photoeye lens.
The Stack Count is short by one.	The Travel Time is too short.	Increase the travel time (see page 57, Travel Time).
The Stack Count is over by one.	The Travel Time is too long.	Decrease the travel time (see page 57, Travel Time).
The interrupt forks do not retract or extend.	The cylinder rod is disconnected from the interrupt fork.	- Remove the interrupt fork guard, reconnect the rod and tighten the jam nut. - If the rod is broken, replace the cylinder.
	Either the cylinder seals or the air fitting connections are leaking.	Repair the leaks and/or replace the worn out cylinder.
	The solenoid valve is not functioning.	- Test the valve by turning the blue manual actuation switch on the end of the valve. - Check the electrical and the air line connections. - Check the output signal from the PLC (the appropriate light should be on when the solenoid valve is energized) see pages 54-55. - Replace the valve.

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The interrupt forks do not retract.	The Count Photoeye is not functioning.	Check the input signal from the photoeye to the PLC. See pages 54-55. Check all electrical connections. The photoeye LED lights up (red) when the photoeye is energized.
Too many pieces are collecting on the interrupt fork, causing an overflow condition.	The Count Photoeye is blocked or dirty.	Clear the obstruction, wipe the lens clean using a soft cloth (See page 34, Photoeyes).
	The Down Time and/or the Stack Up Time are too long.	Adjust the Tower Down Time and/or the Tower Up Time (see pages 57-58).
	The Stacking Tower is not stepping down often enough.	Adjust the Step Count. See Page 56-57.
The Stacking Tower (Stack) is not stepping down.	The Count Photo Eye is blocked or dirty.	Clear the obstruction, wipe the lens clean using a soft cloth. See page 34, Photo Eyes.
	The stacking tower is obstructed.	Clear the obstruction from the stacking tower and or the linear servo shaft.
	The Linear Servo is not functioning.	Refer to Linmot Servo Manual. Information can also be obtained at www.linmot.com .
The Stacking Tower (Tortilla Stack) is stepping down too much.	The Step Count is too low.	Increase the Step Count (see page 56-57).
The Stacking Tower will not return up from the full down position.	There is an obstruction under the stacking towers preventing the tower from lowering to the full down position.	Remove the obstruction from under the tower.
or Stacking Tower won't return up after clear out button is pushed.	The Stack Down proximity switch is not energized.	• Check the proximity switch, the switch must be made before the tower can return up. • Check all the electrical connections. • Check the output signal from the PLC (The corresponding light on the PLC unit will be on when the cylinder is in the full down position) see page 54-55.
The stack is getting tipped over by the stacking tower.	The Down Time is too short.	Increase the Tower Down Time (see page 56, Tower Down Time).
The Stack is out of alignment, side to side.	The Tortilla alignment is out of adjustment.	Adjust the Alignment Guides (see pages 32, Adjustments).

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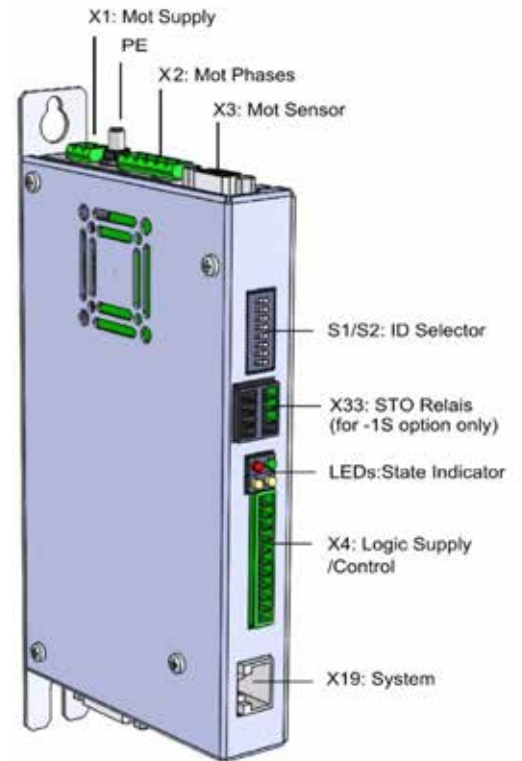
Troubleshooting should be done by qualified personnel only.

Problem	Cause	Solution
The Stack is out of alignment, front to back.	The Infeed / Transfer Conveyor speed is too slow or too fast. The Stacking Platform is out of adjustment.	Adjust the conveyor speed to ensure proper tortilla separation and transfer of the tortilla to the forward position of the stack receiving tray.

Troubleshooting

Servo System - State Indicators

Green	24V Logic Supply Ok
Yellow (lower left)	Motor Enabled / Error Code Low Nibble
Yellow (lower right)	Warning / Error Code High Nibble
Red	Error



Troubleshooting

Servo System - Drive Errors

From the C1100 Servo Drives Installation Guide, page 16

LED Blink Codes			
Error			24VOK
Warn			EN
Error	Warn	EN	Description
Off	Warning	Operation Enabled	Normal Operation: Warnings and operation enabled are displayed.
On	• ~2Hz 0..15 x Error Code High Nibble	• ~2Hz 0..15 x Error Code Low Nibble	Error: The error code is shown by a blink code with "WARN" and "EN". The error byte is divided into low and high nibble (= 4 bit). "WARN" and "EN" are blinking together. The error can be acknowledged. (e.g.: WARN blinks 3x, EN blinks 2x; Error Code = 32h)
• ~2Hz	• ~2Hz 0..15 x Error Code High Nibble	• ~2Hz 0..15 x Error Code Low Nibble	Fatal Error: The error code is shown by a blink code with "WARN" and "EN". The error byte is divided into low and high nibble. "WARN" and "EN" are blinking together. Fatal errors can only be acknowledged by a reset or power cycle. (e.g.: WARN blinks 3x, EN blinks 2x; Error Code = 32h)
• ~4Hz	• ~2Hz 0..15 x Error Code High Nibble	• ~2Hz 0..15 x Error Code Low Nibble	System Error: Please reinstall firmware or contact support.
• ~0.5Hz	• ~0.5Hz	On	Signal Supply 24V too low: The error and warn LEDs blink alternating if the signal supply +24V (X4.2) is less than 18VDC.
Off	○●●●	●○●●	Plug&Play Communication Active: This sequence (Warn on, then En on, then both off, complete sequence of the 4 states ca. 1Sec) signalizes the state when the plug and play parameters are being read from the motor.
○● ~4Hz	●○ ~4Hz	Off	Waiting for Defaulting Parameters: When ID (S1, S2) is set to 0xFF, the drive starts up in a special mode and the Error and Warn LED blink alternating ~4Hz. When the ID is set to 0x00, all parameters will be set to their default value. To leave this state, power down the drive and change the ID. Also see in the Usermanual_LinMot-Talk under chapter trouble shooting.
Off	○● ~2Hz	○● ~2Hz	Defaulting Parameters Done: When the parameters have set to their default values (initiated via S1/S2 on power up) the Warn and En LEDs blink together at 2 Hz. To leave this state, power down the drive. Also see in the Usermanual_LinMot-Talk under chapter trouble shooting.

The meaning of the error codes can be found in the *Usermanual_MotionCtrl_Software_SG5* and the user manual of the installed interface software. These documents are provided together with LinMot-Talk configuration software and can be downloaded from www.linmot.com.

Troubleshooting

Servo System - LinRS Errors

From the LinRS Interface User Manual, page 57

Within the LinRS Intf SW several errors are supported, most of them can be disabled because they are not fatal.

LinRS Error Codes

In the table below the LinRS specific codes are listed

<u>Value</u>	<u>Description</u>
C1h	Checksum error
C2h	Message format error End of Telegram (04h) missing
C3h	Undefined Message Main ID
C4h	Undefined Message Sub ID
C5h	Wrong Baud Rate Defined With S1
C8h	Parameter Unknown UPID
C9h	Parameter Wrong Type

Troubleshooting

Servo System - Motion Errors

From the Motion Control SW User Manual, Pages 109-115

Code	Description	Actions to take
0000h	No Error	No error is pending.
0001h	Err: X4 Logic Supply Too Low	The logic supply voltage has been too low. The minimal logic supply voltage level is defined through parameter 100Eh. Recommended actions: check your 24V logic power supply.
0002h	Err: X4 Logic Supply Too High	The logic supply voltage has been too high. The maximal logic supply voltage level is defined through parameter 1010h. Recommended actions: check your 24V logic power supply.
0003h	Err: X1 Pwr Voltage Too Low	The motor power supply voltage has been too low. The minimal motor supply voltage level is defined through parameter 101Ah. Recommended actions: check your motor power supply, check the wiring, check the sizing of the power supply, add a capacitor too enforce your DC link.
0004h	Err: X1 Pwr Voltage Too High	The motor power supply voltage has been too high. The maximal motor supply voltage level is defined through parameter 101Bh. Back EMF effects may boost the DC link voltage. Recommended actions: check your motor power supply, check the wiring, check the sizing of the power supply, use a regeneration resistor for power dissipation, add a capacitor too enforce your DC link.
0005h	Err: X1 RR Not Connected	A regeneration resistor is configured (see parameter 101Dh) but not connected. Recommended actions: connect the regeneration resistor to X1.
0006h	Err: PTC 1 Sensor Too Hot	The PTC 1 sensor on X4.10 is hot or not connected. Recommended actions: check the temperature, check the wiring
0007h	Err: Min Pos Undershot	The motor position has been below the minimal position (see parameter 146Eh). Recommended actions: check the configuration, check the PLC program
0008h	Err: Max Pos Overshot	The motor position has been above the maximal position (see parameter 146Fh). Recommended actions: check the configuration, check the PLC program
0009h	Err: Ext-Int Sensor Diff Err	The position difference between sensor feedback on X3 and sensor feedback on X12 has been too big. Recommended actions: check sensor wiring, check sensor configuration (count direction, etc.), check parameter 1266h
000Ah	Fatal Err: X12 Signals Missing	The external sensor is not connected to X12 or the wiring is not ok. Recommended actions: check the wiring
000Bh	Err: Pos Lag Always Too Big	The motor was not able to follow the demand position. The maximal allowed position difference is defined through parameter 1473h. Recommended actions: check the motor load, check the motor stroke range for possible collisions, check the position controller setup, check the set-point generation (unreachable speed/acceleration values?), check the motor sizing.

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Servo System - Motion Errors

From the Motion Control SW User Manual, Pages 109-115

Code	Description	Actions to take
000Ch	Err: Pos Lag Standing Too Big (Not on B1100)	The motor was not able to reach the target position or was not able to stay at the target position. The maximal allowed position difference is defined through parameter 1475h. Recommended actions: check the motor load, check the motor stroke range for possible collisions, check the position controller setup, check the motor sizing
000Dh	Fatal Err: X1 Pwr Over Current	Over current on X1 detected. Recommended actions: check motor wiring, check motor configuration, for P01-48 type motors: set parameter 11F4h to value 0001h overload also possible due wrong motion profile or too much friction.
000Eh	Err: Supply Dig Out Missing	Controller board defective. Recommended actions: contact support for repair
000Fh	Err: PTC 2 Sensor Too Hot	The PTC 2 sensor on X4.11 is hot or not connected. Recommended actions: check the temperature, check the wiring
0010h	Err: Controller Ph1+ Too Hot	Servo controller power bridge phase 1+ too hot. Recommended actions: check motor wiring
0011h	Err: Controller Ph1- Too Hot	Servo controller power bridge phase 1- too hot. Recommended actions: check motor wiring
0012h	Err: Controller Ph2+ Too Hot	Servo controller power bridge phase 2+ too hot. Recommended actions: check motor wiring
0013h	Err: Controller Ph2- Too Hot	Servo controller power bridge phase 2- too hot. Recommended actions: check motor wiring
0014h	Err: Controller Pwr Too Hot	DC link temp sensor has detected over temperature. Recommended actions: check wiring
0015h	Err: Controller RR Hot Calc	Regeneration resistor switch hot. Recommended actions: check RR configuration (Turn On level, Resistance, etc.), check RR sizing
0016h	Err: Controller X3 Too Hot	Temp sensor on X3 has detected over temperature. Recommended actions: check motor wiring
0017h	Err: Controller Core Too Hot	Temp sensor on controller's PCB board reports core being hot.
0018h	Err: Power Bridge Ph1+ Defective	Servo controller power bridge phase 1+ may be defective. Recommended actions: contact support
0019h	Err: Power Bridge Ph1- Defective	Servo controller power bridge phase 1- may be defective. Recommended actions: contact support
001Ah	Err: Power Bridge Ph2+ Defective	Servo controller power bridge phase 2+ may be defective. Recommended actions: contact support
001Bh	Err: Power Bridge Ph2- Defective	Servo controller power bridge phase 2- may be defective. Recommended actions: contact support
001Ch	Err: Supply DigOut X6 Fuse Blown	Supply fuse for digital outputs on X6 blown. Recommended actions: check X6 wiring, contact support for repair

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Servo System - Motion Errors

From the Motion Control SW User Manual, Pages 109-115

Code	Description	Actions to take
001Dh	Err: Supply X3.3 5V Fuse Blown	Supply X3.3 5V fuse blown. Motor or and/or wiring defective. Recommended actions: contact support for controller repair, check motor and wiring, replace motor and motor cables
001Eh	Err: Supply X3.8 AGND Fuse Blown	Supply X3.8 analog ground fuse blown. Recommended actions: contact support for controller repair, check motor and wiring, replace motor and motor cables
0020h	Err: Motor Hot Sensor	Temp sensor reports hot motor. Recommended actions: wait until motor has cooled down (until corresponding warning disappears, check load, check the motor configuration, check the setpoint generation (unreachable speed/acceleration values?), check the motor sizing
0021h	Fatal Err: X3 Hall Sig Missing	Motor hall signals not connected to X3 or motor defective: Recommended actions: Power down the controller and all power supplies, then reconnect motor, check motor and wiring, check parameter 1221h.
0022h	Fatal Err: Motor Slider Missing	Motor hall sensors cannot see magnetic field of the slider. The motor position was outside the allowed range defined through the motors ZP and Max Stroke data (see data sheet). Recommended actions: check stroke range, check slider orientation.
0023h	Err: Motor Short Time Overload	Short time motor overload detected. Recommended actions: check if motor is blocked, check motor sizing
0024h	Err: RR Hot Calculated	Regeneration resistor hot calculated. Recommended actions: check RR configuration (Turn On level, Resistance, etc.), check RR sizing
0025h	Err: Sensor Alarm	Sensor Alarm On X12 Occurred. Recommended actions: Check sensor mounting, band contamination or motion speed
0028h	Err: Ph1+ Short Circuit To GND	Short circuit between phase 1+ and ground detected. Recommended actions: check motor wiring, check motor
0029h	Err: Ph1- Short Circuit To GND	Short circuit between phase 1- and ground detected. Recommended actions: check motor wiring, check motor
002Ah	Err: Ph2+ Short Circuit To GND	Short circuit between phase 2+ and ground detected. Recommended actions: check motor wiring, check motor
002Bh	Err: Ph2- Short Circuit To GND	Short circuit between phase 2- and ground detected. Recommended actions: check motor wiring, check motor
002Ch	Err: Ph1 Short Circuit To Ph2	Short circuit between motor phase 1 and phase 2 detected. Recommended actions: check motor wiring, check motor
0030h	Err: Ph1+ Wired To Ph2+	Motor phase 1+ has contact to phase 2+. Recommended actions: check motor wiring, check motor
0031h	Err: Ph1+ Wired To Ph2-	Motor phase 1+ has contact to phase 2-. Recommended actions: check motor wiring, check motor
0032h	Err: Ph1+ Not Wired To Ph1-	Motor phase 1+ has no connection to phase 1-. Recommended actions: check motor wiring, check motor

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Servo System - Motion Errors

From the Motion Control SW User Manual, Pages 109-115

Code	Description	Actions to take
0033h	Err: Ph2+ Wired To Ph1+	Motor phase 2+ has contact to phase 1+. Recommended actions: check motor wiring, check motor
0034h	Err: Ph2+ Wired To Ph1-	Motor phase 2+ has contact to phase 1-. Recommended actions: check motor wiring, check motor
0035h	Err: Ph2+ Not Wired To Ph2-	Motor phase 2+ has no connection to phase 2-. Recommended actions: check motor wiring, check motor
0036h	Err: Ph1 Short Circuit To Ph2+	Short circuit between motor phase 1 and phase 2+ detected. Recommended actions: check motor wiring, check motor
0037h	Err: Ph1 Short Circuit To Ph2-	Short circuit between motor phase 1 and phase 2- detected. Recommended actions: check motor wiring, check motor
0038h	Err: Ph2 Short Circuit To Ph1+	Short circuit between motor phase 2 and phase 1+ detected. Recommended actions: check motor wiring, check motor
0039h	Err: Ph2 Short Circuit To Ph1-	Short circuit between motor phase 2 and phase 1- detected. Recommended actions: check motor wiring, check motor
003Ah	Err: Phase U Broken	Motor phase U broken. Recommended actions: check motor wiring, check motor
003Bh	Err: Phase V Broken	Motor phase V broken. Recommended actions: check motor wiring, check motor
003Ch	Err: Phase W Broken	Motor phase W broken. Recommended actions: check motor wiring, check motor
0040h	Err: X4.3 Brake Driver Error	X4.3 brake driver reports error. Recommended actions: check for short circuit on X4.3
0041h	Err: Dig Out X4.4..X4.11 Status	X4.3..X4.11 output driver reports error. Recommended actions: check for short circuit on outputs X4.4..X4.11 or output configurations.
0042h	Err: Dig Out X6 Status	X6 output driver reports error. Recommended actions: check for short circuit on outputs X6.
0044h	Err: X4 Dig Out GND Fuse Blown	Ground fuse for digital outputs on X4 blown. Recommended actions: check X4 wiring, contact support for repair
0045h	Fatal Err: Motor Comm Lost	Motor communication lost. Recommended actions: Power down and check motor wiring and motor, replace cable and/or motor.
0046h	Err: PTC 1 Broken	PTC 1 on X4.10 broken or not connected. Recommended actions: Power down and check PTC 1 wiring and resistance.
0047h	Err: PTC 1 Short To 24V	PTC 1 on X4.10 short to 24V. Recommended actions: Power down and check PTC 1 wiring and resistance.
0050h	Setup Err: HW Not Supported	Setup error, hardware is not supported by the software. Recommended actions: download correct firmware, contact support

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Servo System - Motion Errors

From the Motion Control SW User Manual, Pages 109-115

Code	Description	Actions to take
0051h	Setup Err: SW Key Missing	Software key and access code for special functionality is missing. Recommended actions: Order the SW key with your support together with the serial number of your HW.
0058h	Runtime Err: ROM write error	Runtime error, MC SW was not able to change parameter value in ROM. Recommended actions: verify PLC is not configuring during this action, contact support
0060h	Cfg Err: RR Voltage Set Too Low	Configuration error: regeneration resistor turn on/off voltage parameter value is too low. Recommended actions: check parameters 101Eh and 101Fh
0061h	Cfg Err: RR Hysteresis < 0.5V	Configuration error: regeneration resistor turn on/off voltage parameter values too close to each other. Recommended actions: check parameters 101Eh and 101Fh
0062h	Cfg Err: Curve Not Defined	Configuration error. Software tried to start a curve that is not defined yet. Action to take: define the curve using the curves service, check if curves were downloaded to controller, check the curve IDs, check the configuration, check the PLC program
0063h	Cfg Err: Pos Ctrl Max Curr High	Configuration error: Invalid max current setting in control parameters. Recommended actions: check parameters 13A6h and 13BAh, check PLC program
0064h	Cfg Err (Fatal): No Motor Defined	Configuration error: No motor has been configured yet. Recommended actions: use the motor wizard to configure the motor
0065h	Cfg Err (Fatal): No Trigger Mode Defined	Configuration error: Digital input X4.6 is configured for trigger input function, but the trigger mode is not defined yet. Recommended actions: configure parameter 170Ch
0067h	Cfg Err (Fatal): Wrong Stator Type	Configuration error: The configured motor type does not match with the connected motor. Recommended actions: configure correct motor type by using the motor wizard, connect an appropriate motor. If a PnP motor is connected, press the DEF button to default the parameters and reboot the drive.
0068h	Cfg Err (Fatal): No Motor Communication	Configuration error: The controller was not able to establish the communication to the microcontroller on the motor. Older P01 motors don't support motor communication. Recommended actions: check motor wiring, check motor, check the motor configuration, disable communication by using parameter 11FBh if you have an old P01 motor.
0069h	Cfg Err: Wrong Slider	Configuration error: A wrong slider has been configured or slider home position has an invalid value. Recommended actions: reconfigure the motor by using the motor wizard
0080h	User Err: Lin: Not Homed	User error: The PLC program tried to start an action that requires the motor to be already homed, but the motor was not homed. Recommended actions: check the PLC program, do a homing of the motor first

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Servo System - Motion Errors

From the Motion Control SW User Manual, Pages 109-115

Code	Description	Actions to take
0081h	User Err: Unknown Motion Cmd	User error: The PLC program sent an unknown motion command ID. Recommended actions: check PLC program, check firmware version
0082h	User Err: PVT Buffer Overflow	User error: The PLC program has sent the stream position commands too fast, the buffer had an overflow. Streaming has to be strictly cyclic! Recommended actions: check PLC program, check the fieldbus by using bus monitor tools
0083h	User Err: PVT Buffer Underflow	User error: The PLC program has sent the stream position commands too slowly, the buffer had an underflow. Streaming has to be strictly cyclic! Recommended actions: check PLC program, check the fieldbus by using bus monitor tools
0084h	User Err: PVT Master Too Fast	User error: The PLC program has begun to send PVT streaming command. The commands were too close to each other. The servo controller expects new streaming commands every 2ms to 5ms. Recommended actions: check PLC program, check the fieldbus by using bus monitor tools
0085h	User Err: PVT Master Too Slow	User error: The PLC program has begun to send PVT streaming command. The cycle time between the streaming commands has been too long. The servo controller expects new streaming commands every 2ms to 5ms. Recommended actions: check PLC program, check the fieldbus by using bus monitor tools
0086h	User Err: Motion Cmd In Wrong St	User error: The PLC program has sent a motion command while the controller was not in an appropriate operational state. Most of the motion commands are accepted only in operational state 8 (Operation Enabled). Recommended actions: check the PLC program
0087h	User Err: Limit Switch In High	User error: The motor moved into the Limit Switch In while it was still in the stroke range. Recommended actions: check the PLC program or check homing
0088h	User Err: Limit Switch Out High	User error: The motor moved into the Limit Switch Out while it was still in the stroke range. Recommended actions: check the PLC program or check homing
0089h	User Err: Curve Amp Scale Error	User error: The automatic calculated amplitude scale is out of range -2000percent to 2000percent. Recommended actions: check the PLC program or use other curve
008Ah	User Err: Cmd Tab Entry Not Def	Called command Table entry is not defined. Recommended actions: check the PLC program or define Command Table Entry.

Troubleshooting

Linmot Stator Troubleshooting

Linmot stators should be tested when a bad stator is suspected. Also, when replacing a bad drive, all stators connected to the drive should be tested to prevent possible damage to the new drive.

The table shows the resistive values between the connector pins. If the value is not in a range of +/- 10%, the stator should be considered damaged and should be replaced. If the stator passes the resistance check, it may be connected to the drive and then tested for operation. Note that it is possible for a stator to pass the resistance check, but may not work when connected to a drive.

PS01-37x240-C (0150-1224)

Phase 1+ / Phase 1-	Pin A / Pin B	11.5Ω
Phase 2+ / Phase 2-	Pin C / Pin D	11.5Ω
5V / GND	Pin E / Pin F	155Ω / 275Ω*
Sensor Sinus / GND	Pin G / Pin F	33 kΩ / 40 kΩ*
Sensor Cosine / GND	Pin H / Pin F	33 kΩ / 40 kΩ*
Temp. Sensor / GND	Pin L / Pin F	10kΩ / >20MΩ*
Phase / GND	Pin A,B,C,D / Pin F	>20 MΩ
All Pin / Shield	Pin A-L / Housing	>20 MΩ

* For stators with serial no. before xxx.38A.xxx

